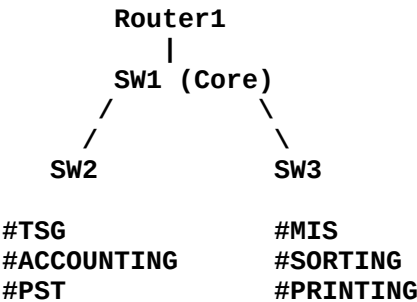


3-switch hierarchical VLAN

topology where:

- SW1 is the Core Switch.
- SW2 and SW3 are Access Switches.
- Each switch hosts 2 departments (VLANs).
- The router performs **inter-VLAN routing** using Router-on-a-Stick.

Topology



VLAN Plan

VLAN ID	Department	Network	Gateway
10	TSG	192.168.10.0/24	192.168.10.1
20	ACCOUNTING	192.168.20.0/24	192.168.20.1
30	PST	192.168.30.0/24	192.168.30.1
40	MIS	192.168.40.0/24	192.168.40.1
50	SORTING	192.168.50.0/24	192.168.50.1
60	PRINTING	192.168.60.0/24	192.168.60.1

SW1 (Core Switch)

Create VLANs

```
enable
conf t

vlan 10
  name TSG

vlan 20
  name ACCOUNTING

vlan 30
  name PST

vlan 40
  name MIS
```

```
vlan 50
  name SORTING

vlan 60
  name PRINTING
```

Department Ports

```
interface fa0/1
  switchport mode access
  switchport access vlan 10

interface fa0/2
  switchport mode access
  switchport access vlan 20
```

Trunk Ports

```
interface gi0/1
  switchport mode trunk

interface gi0/2
  switchport mode trunk

interface gi0/3
  switchport mode trunk
Gi0/1 -> Router1
Gi0/2 -> SW2
Gi0/3 -> SW3
```

SW2

Create VLANs

```
vlan 10
vlan 20
vlan 30
vlan 40
vlan 50
vlan 60
```

Department Ports

```
interface fa0/1
  switchport mode access
  switchport access vlan 30

interface fa0/2
  switchport mode access
  switchport access vlan 40
```

Trunk to Core

```
interface gi0/1
  switchport mode trunk
```

SW3

Create VLANs

```
vlan 10
vlan 20
vlan 30
vlan 40
vlan 50
vlan 60
```

Department Ports

```
interface fa0/1
  switchport mode access
  switchport access vlan 50

interface fa0/2
  switchport mode access
  switchport access vlan 60
```

Trunk to Core

```
interface gi0/1
  switchport mode trunk
```

Router1 Configuration

```
enable
conf t
```

```
interface g0/0/0
  no shutdown
```

```
interface g0/0/0.10
  encapsulation dot1Q 10
  ip address 192.168.10.1 255.255.255.0
```

```
interface g0/0/0.20
  encapsulation dot1Q 20
  ip address 192.168.20.1 255.255.255.0
```

```
interface g0/0/0.30
  encapsulation dot1Q 30
  ip address 192.168.30.1 255.255.255.0
```

```
interface g0/0/0.40
  encapsulation dot1Q 40
  ip address 192.168.40.1 255.255.255.0
```

```
interface g0/0/0.50
  encapsulation dot1Q 50
  ip address 192.168.50.1 255.255.255.0
```

```
interface g0/0/0.60
  encapsulation dot1Q 60
  ip address 192.168.60.1 255.255.255.0
```

Example PC Addressing

SW1

TSG PC
IP: 192.168.10.10
GW: 192.168.10.1

ACCOUNTING PC
IP: 192.168.20.10
GW: 192.168.20.1

SW2

PST PC
IP: 192.168.30.10
GW: 192.168.30.1

MIS PC
IP: 192.168.40.10
GW: 192.168.40.1

SW3

SORTING PC
IP: 192.168.50.10
GW: 192.168.50.1

PRINTING PC
IP: 192.168.60.10
GW: 192.168.60.1

Verification

```
show vlan brief
show interfaces trunk
show ip interface brief
```

Expected trunk links:

SW1 Gi0/1 -> Router
SW1 Gi0/2 -> SW2
SW1 Gi0/3 -> SW3
SW2 Gi0/1 -> SW1
SW3 Gi0/1 -> SW1

Expected result:

- All 6 VLANs are carried across trunks using **802.1Q**.
- Each department is isolated in its own VLAN.
- The router allows communication between departments.
- PCs in TSG, ACCOUNTING, PST, MIS, SORTING, and PRINTING can all reach each other through Router1.